Learning Collapse Detection from Student Activity-Patterns in Large-Scale Educational Data

Final Review Report

Course Name: Big Data Analytics

VIT®
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1. Abstract:

Universities and online learning platforms collect a lot of information about how students learn, such as how often they log in, watch lectures, complete assignments, and participate in activities. Most current systems use this data only to predict final results like grades or whether a student may drop out. But this approach misses an important problem. Students usually do not fail suddenly. Instead, their learning slowly becomes weaker over time. They may still be studying and putting in effort, but they stop improving, lose focus, or repeat the same mistakes. This slow decline in learning is called learning collapse.

This project focuses on finding these early warning signs before the student actually performs badly. Instead of looking only at final results, it studies how student behavior changes over time. For example, it checks if a student's activity becomes irregular, if their participation decreases, or if their effort is no longer effective. To do this, the study uses large datasets like the Open University Learning Analytics Dataset, which contain detailed records of student activities in online courses.

Since this data is very large, special technologies are needed to store and process it efficiently. Tools like Apache Hadoop help store huge amounts of data, while Apache Spark allows fast analysis, and Apache Hive helps in querying and organizing the data. The project also uses simple data analysis methods like finding patterns in how students behave over time and grouping students with similar learning habits.

Overall, this study shows that current systems do not focus enough on early learning problems. It aims to build a system that can detect learning collapse early, so teachers and institutions can take action in time and help students improve before their performance gets worse.

2. Introduction & Literature Survey:

Learning Collapse Detection is the process of identifying the stage where a student's learning starts to become unstable or ineffective, even though the student is still putting in effort. In many cases, students continue studying, attending classes, and completing tasks, but their learning stops improving or becomes inconsistent. This condition is called learning collapse because the learning process is no longer productive. It can be seen through patterns such as irregular study habits, sudden changes in activity levels, reduced consistency in engagement, and frequent ups and downs in performance.

These changes are not always visible through final grades or results, but they can be observed by analyzing large amounts of student activity data collected from online learning systems. For example, a student might log in regularly but spend less meaningful time on learning materials, or their participation may become uneven over time. Such behavioral fluctuations indicate that the student is struggling, even if they have not yet failed academically.

Detecting learning collapse at an early stage is very important because it happens before clear signs of failure appear. If identified in time, teachers and institutions can take action, such as providing extra support, guidance, or personalized learning strategies. By using data-driven

methods and analyzing patterns in student behavior, this approach helps in understanding hidden learning problems and allows timely intervention, ultimately improving student success and preventing future academic failure.

Literature Survey:

Sr No.	Paper Title (Year of Publication)	Proposed System	Features (Pros)	Limitations (Cons)	Research Gap Analysis
1	Applying Machine Learning on Big Data With Apache Spark (2025)	Evaluates multiple ML models (LinR, LR, RF, GBT, SVM, SVR, KNN) using Apache Spark MLlib on large datasets for regression and classification.	- High scalability using Spark's Distributed in-memory processing. - Comprehensive comparison of models and resource usage	Focuses on general big data benchmarks, not education specific datasets	Does not analyze student behavior datasets (e.g., OULAD) or early learning collapse prediction in educational systems
2	Integration of Data Mining Clustering Approach in the Personalized E-Learning System (2018)	A personalized e-learning architecture using advanced clustering (CFSFDP-HD) to group students based on learning behavior	- Adaptive clustering without predefined cluster count - Handles noisy educational big data effectively	- Tested only on simulated datasets - Not evaluated on real large scale LMS logs	Lacks validation on real-world large-scale student activity data and does not address early learning collapse detection over time
3	Integration of Data Mining Clustering Approach in the Personalized E-Learning System (2018)	A personalized e-learning architecture integrating educational data mining clustering (CFSFDP-HD) to group students based on learning behavior.	- Adaptive clustering without predefined cluster count - Handles noisy and complex educational data effectively	- Evaluated only on simulated datasets - No real-time or large-scale deployment validation	Lacks real-world large-scale implementation and integration with distributed big data platforms like Spark/Hadoop for continuous learning collapse detection.
4	Big Data and Analytics in Education: Leveraging Data to Improve Student Performance	Uses Hadoop-based big data analytics with MapReduce, VADER sentiment analysis, and collaborative filtering to analyze student	- Faster processing of large educational datasets - Supports personalization and sentiment-based	- Lacks real-world large-scale deployment validation - Limited focus on predictive	- Does not model early risk or learning collapse patterns using longitudinal student activity data

	(2025)	performance and behavior	insights	learning collapse detection	
5	Data Mining and Machine Learning Applications for Educational Big Data in the University (2019)	Uses educational big data (grades, attendance, GPA) with ML models to predict dropout, graduation delay, poor performance, and recommend courses	- Integrates multiple academic features - Applies diverse ML models (NN, RF, SVM, NB) - Early risk prediction for students	- Uses institution-specific data - Limited scalability discussion - No distributed big data processing	Does not leverage large-scale distributed frameworks (Hadoop/Spark) or real-time big data pipelines for scalable learning collapse detection
6	Big Data Machine Learning using Apache Spark MLlib (2017)	Uses Apache Spark MLlib 2.0 for distributed machine learning on massive datasets and compares performance with Weka on Hadoop.	- Highly scalable and fast due to in-memory distributed processing - Supports multiple ML algorithms (classification, clustering, regression)	- Focuses mainly on computational performance, not predictive insight - Does not address learning behavior or educational datasets	Lacks application to educational big data and does not analyze student learning patterns or early risk detection, creating scope for learning-collapse prediction using academic datasets like OULAD.
7	Big Data for Institutional Planning, Decision Support and Academic Excellence (2016)	Hadoop-based big data architecture for educational decision support using institutional, sensor, and feedback data	- Improves institutional planning and academic decision-making using big data analytics	- Conceptual architecture only - no real dataset or predictive model validation	- Lacks student-level predictive modeling and learning-collapse detection using real educational datasets
8	Big Data and Analytics in Education: Leveraging Data to Improve Student Performance (2025)	Uses Hadoop-based Big Data framework integrating MapReduce, VADER sentiment analysis, and Collaborative Filtering to analyze student performance and	- Scalable processing of structured & unstructured educational data - Supports personalized learning and faster analytics than RDBMS	- Focuses mainly on analytics efficiency rather than predictive learning failure - Limited real-time intervention	Does not model early learning collapse detection or predictive dropout risk using longitudinal student activity data

		feedback.		mechanisms	
9	Using Big Data and Hadoop in the Student Learning Process – Enhancing the Educational Process through Real Experience (2023)	Uses Hadoop ecosystem with Hive and Hue to provide hands-on big data learning experience for students.	- Enhances practical skills using real datasets and Hadoop tools	- No quantitative evaluation of learning outcomes or performance gains	Does not apply learning analytics or predictive models to analyze student academic performance.
10	Research on the Process Data Mining of Online Wisdom Learning (2019)	Big-data-driven framework for online learning process evaluation using behavioral and cognitive learning data	- Supports dynamic, real-time process evaluation - Integrates motivation, behavior, and learning outcomes	- Lacks implementation details and real system validation	No predictive or scalable learning-collapse detection model using real large-scale datasets like OULAD
11	ML-Based Multidimensional Big Data Analytics over Clouds (2023)	ML-driven multidimensional analytics using compressed OLAP cubes on cloud	- High scalability and efficient analytics over massive data	- Complex architecture and high computational cost	Not applied to educational datasets or learning behavior analytics
12	An Insight into Tree-Based Machine Learning Techniques for Big Data Analytics Using Apache Spark (2017)	Spark-based implementation of tree-based ML models (Decision Tree, Random Forest, Gradient Boosted Trees) replacing MapReduce for big data prediction	- Faster execution and lower disk I/O with improved time and space efficiency compared to MapReduce	- Evaluated on limited datasets and focused mainly on tree-based models	Lacks evaluation on diverse ML models, larger heterogeneous datasets, and real-time streaming analytics
13	Big Data Platform for Educational Analytics (2021)	Cloud-based educational big data platform using data lake architecture with batch and real-time analytics	- End-to-end scalable platform - Supports non-data scientists via tools like Tableau	- No deep learning or adaptive models - Platform validation limited to few case studies	Lacks predictive, personalized learning models and real-time intervention mechanisms for student behavior analysis

14	Open University Learning Analytics Dataset (2017)	Development of a centralized educational dataset for learning analytics research. Collection of student demographics, assessment scores, and VLE interaction logs.	- Large-scale real student data - Rich temporal logs (VLE clicks, assessments) - Widely accepted benchmark - Supports longitudinal analysis	- No embedded predictive model - Static dataset structure - Limited behavioral interpretation	Dataset is mostly used for final outcome prediction; lacks work on early learning instability, entropy, or collapse dynamics
15	Predicting Student Performance Using Machine Learning (2018)	Application of traditional ML models for grade prediction. Feature extraction from assessment and demographic data. Classification of students into performance categories	- Uses classification & regression models - Early prediction capability - Simple feature engineering	- Focus on accuracy only - Ignores temporal evolution - Assumes steady learning	Does not capture learning behavior fluctuations or pre-failure degradation patterns
16	MOOC Dropout Prediction Using Deep Learning (2019)	Deep neural networks trained on MOOC interaction data. Automated feature learning from logs. Binary classification of dropout vs non-dropout	- Handles large-scale data - High predictive accuracy - Automated feature learning	- Black-box behavior - Dropout ≠ learning collapse - Late-stage prediction	Dropout is treated as an endpoint; collapse onset timing and causes remain unexplored
17	Knowledge Tracing Models for Student Assessment (2020)	Sequential modeling of student concept mastery. Probability estimation of skill acquisition. Adaptive assessment support	- Fine-grained skill tracking - Sequential modeling - Adaptive assessment support	- Assumes monotonic mastery growth - Poor handling of regression	Sudden learning collapse, forgetting spikes, and stress-induced failures are ignored
18	Early Warning Systems in Learning Analytics (2021)	Rule-based monitoring of student activity and grades. Instructor alert generation for at-risk students.	- Actionable dashboards - Real-time monitoring - Institutional adoption	- Rule-based thresholds - Reactive alerts	Lack of predictive collapse indicators before grades decline

		Dashboard-driven academic monitoring			
19	Temporal Learning Analytics Using Clickstream Data (2022)	Time-series analysis of LMS interaction logs. Sequential pattern mining of learning activities. Visualization of temporal engagement trends	- Captures sequential behavior - High-resolution activity logs	- Noisy data -Computationally heavy	Temporal data is used descriptively, not to compute learning stability or entropy metrics
20	Early Prediction of Academic Failure Using Behavioral Data (2020)	Design of a behavioral risk scoring framework for academic failure prediction. Integration of assessment scores with LMS interaction and activity logs.	- Combines assessment performance and behavioral data - Enables early-stage academic risk identification	- Linear assumptions - No dynamics	Cannot detect non-linear collapse trajectories
21	Automatic Generation of Training Data for Anomaly Detection Using Other User's Data Samples (IBM, 2018)	Automatically creates training data for anomaly detection by comparing data from different users using LOF.	- Reduces manual data labeling - Simple anomaly detection approach	- Works only when user behavior is similar - Not suitable for real-time big data	Does not support Hadoop or large-scale distributed processing for diverse user data
22	Ambulatory System for Measuring Physical Activity and Automatic Fall Detection (2012)	Uses wearable sensor data to detect physical activity and falls.	- Simple sensor-based system - Real-time fall detection	- Big data storage not discussed - Not scalable	Does not use big data platforms like Hadoop for storing and analyzing large sensor data

23	Fall Detection and Fall Risk Detection Systems and Methods (2016)	Detects falls using motion and posture patterns from wearable sensors.	- Improved fall detection accuracy - Uses posture analysis	- Large data processing not explained	No support for big data analytics tools to handle large sensor datasets
24	Detection of Student Engagement in E-Learning Environments Using Deep Learning (2024)	Uses deep learning models on video data to detect student engagement.	- Automatic engagement detection - Uses temporal video analysis	- Needs high computing power - Limited accuracy	Does not use Hadoop or distributed systems for large video data processing
25	Student's Engagement Detection Based on Computer Vision: A Review (2025)	Reviews multiple computer vision and deep learning methods for engagement detection.	- Covers many existing methods Identifies challenges	- No new system proposed	Highlights lack of scalable big data solutions and real-time processing frameworks
26	Data Retrieval for Anomaly Detection (IBM, 2022)	Controls how much data is collected and how often for anomaly detection.	- Efficient data collection - Reduces system load	- Focuses only on data collection	Does not explain how Hadoop or big data analytics can be used for detection.

3. Problem Statement:

There is a lack of scalable analytical frameworks that detect early learning collapse by analyzing temporal student activity patterns in large-scale educational data.

Problem Statement Explanation:

1. Existing educational analytics systems primarily focus on predicting final academic outcomes such as grades or dropout, rather than analyzing the learning behavior that precedes failure.
2. Student learning often degrades gradually due to saturation, stress, or inconsistent engagement, which is not captured by traditional performance-based models.
3. Most data mining approaches assume steady learning progression and ignore temporal instability in student activity patterns.
4. Large-scale student activity logs remain underutilized for detecting early learning collapse using behavioral indicators.
5. There is a need for a scalable analytical approach that models learning instability using temporal student activity data before academic failure occurs.

4. Methodology & Design:

The methodology of this project focuses on detecting learning collapse by analyzing student activity data step by step. First, student interaction data is collected from large datasets such as the Open University Learning Analytics Dataset, which contains details like clicks, course access, and learning behavior.

This data is then stored in a distributed system using Hadoop HDFS to handle large volumes efficiently. After storage, the data is processed using Apache Spark, where it is cleaned and organized for analysis. In the next step, feature engineering is performed to convert raw data into meaningful indicators such as total clicks, number of active days, and variation in activity, which help in understanding student engagement and learning stability. These features are then used in a machine learning technique called K-Means clustering to group students based on their behavior patterns into categories like active, moderate, and low engagement students. Students in the low engagement group are considered at risk of learning collapse.

Finally, the results are analyzed and visualized to identify patterns and provide insights, which can help educators take early action and support students before their performance declines.

System Design:

The system design shows how different components work together to detect learning collapse.

The process starts with **data sources**, which include student activity logs and datasets like OULAD. These contain detailed information about student interactions.

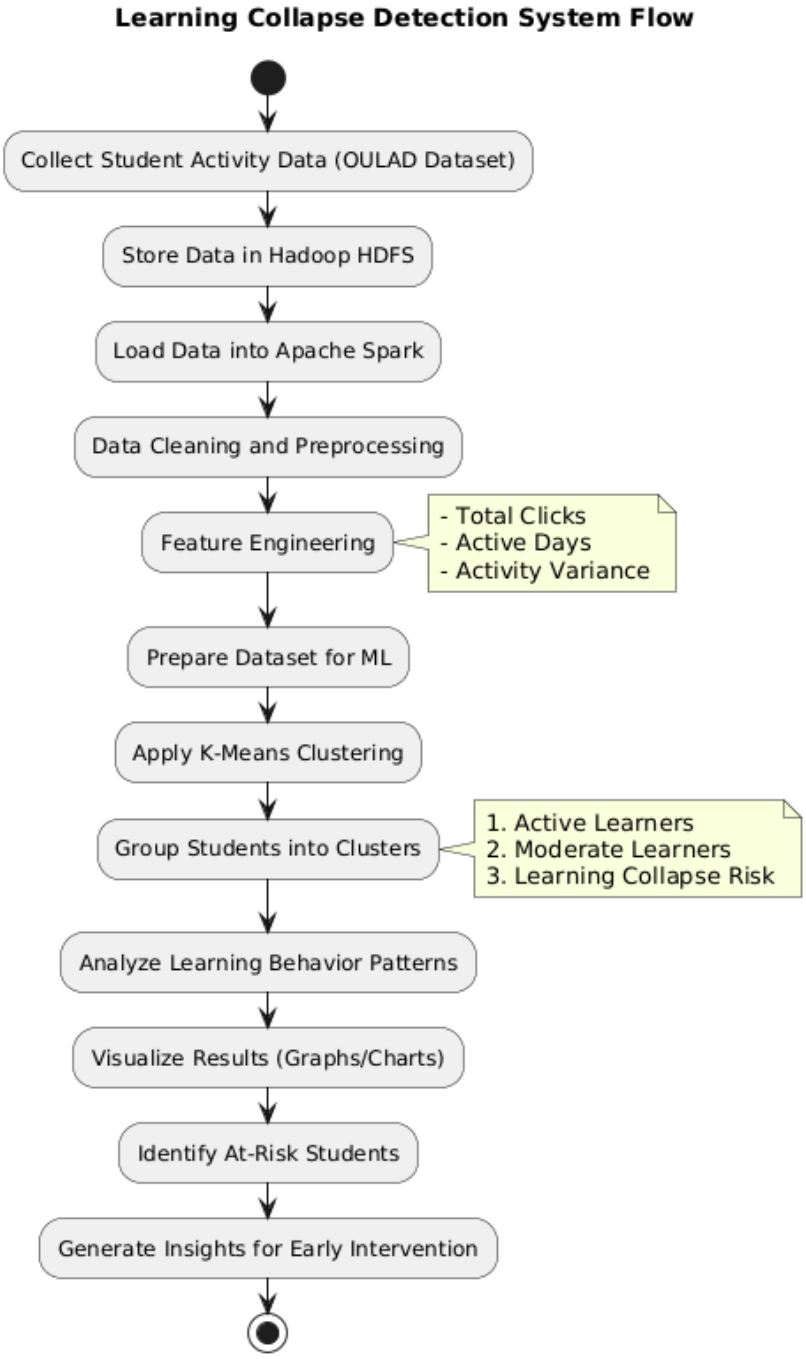
The data is then sent to **Hadoop HDFS**, where it is stored in a distributed manner. This ensures that even very large datasets can be handled efficiently.

Next, **Apache Spark** processes the data. It performs data cleaning, transformation, and feature extraction. Spark is used because it is fast and can process big data in parallel.

After processing, the system uses **machine learning (K-Means clustering)** to group

students based on their learning behavior. This step is important because it helps identify patterns in student engagement.
Once the analysis is complete, **visualization tools** like graphs are used to display the results clearly. Teachers and institutions can use these insights to identify students who are at risk.

Flow diagram:

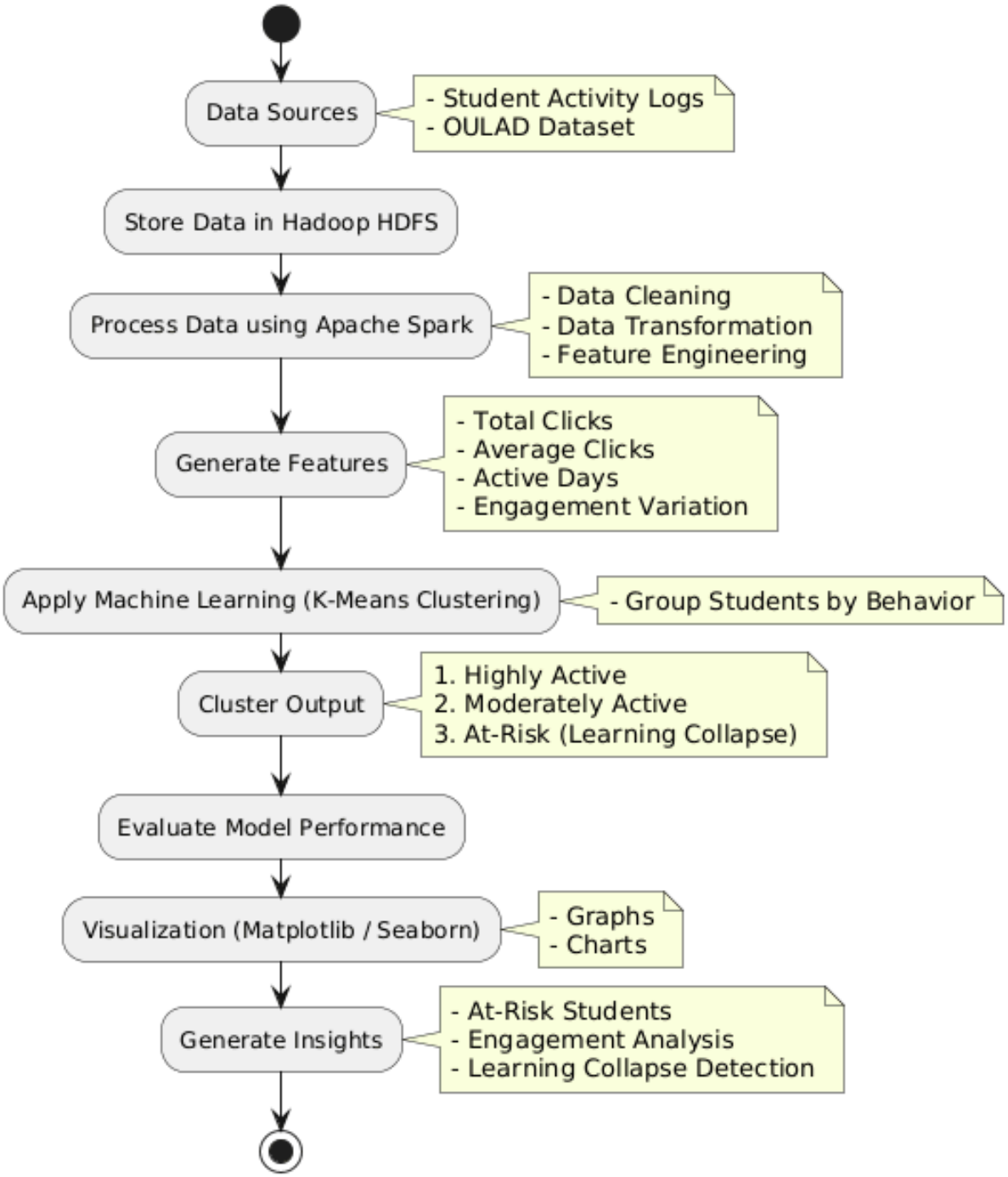


System architecture:

The system architecture explains how student activity data moves through different big data technologies to detect learning collapse. The process starts with data sources such as student activity logs and the Open University Learning Analytics Dataset. These datasets contain information about how students interact with the learning platform, including clicks, course access, and assessment activities.
The raw data is first stored in Hadoop Distributed File System (HDFS), which allows large amounts of data to be stored across multiple systems, making it scalable and reliable. After storing the data, it is processed using Apache Spark. Spark is used to clean the data, transform it, and perform feature engineering. In this stage, important features such as total clicks, average clicks, active days, and engagement variation are created from the raw data.

The processed data is then used for machine learning analysis. In this project, the K-Means clustering algorithm is applied using Spark MLlib to group students based on their learning behavior. The students are categorized into groups like highly active learners, moderately active learners, and students at risk of learning collapse.
After clustering, the results are evaluated to understand how well the model identifies different learning patterns. Then, Python visualization tools such as graphs and charts are used to represent student engagement and cluster distribution in an easy-to-understand way.
Finally, the system generates useful insights such as identifying at-risk students, analyzing engagement patterns, and detecting early learning collapse. These insights help educators monitor student performance and take timely action to support students before their learning declines.

Learning Collapse Detection System Architecture



5. Implementation Steps:

Dataset Description:

The dataset used in this project is the Open University Learning Analytics Dataset (OULAD).This dataset contains detailed information about student activity in an online learning environment.

It includes data such as:

1. student demographic information
2. course registrations
3. assessment results
4. student interaction with learning materials

The most important dataset used in this project is **studentVle**, which contains clickstream data showing how students interact with course materials.

Each record in the dataset contains:

1. code_module → course module
2. code_presentation → course semester
3. id_student → student ID
4. id_site → learning resource accessed
5. date → interaction date
6. sum_click → number of clicks

By analyzing the number of clicks and interaction patterns, we can understand **student engagement behavior**.

Students who gradually reduce their activity may be experiencing **learning collapse**, which means their learning effectiveness is decreasing.

```
>>> df = spark.read.csv("/home/arya_mulay/oulad/studentVle.csv", header=True, inferSchema=True)
>>> df.show(5)
+-----------+-----------------+----------+-------+----+---------+
|code_module|code_presentation|id_student|id_site|date|sum_click|
+-----------+-----------------+----------+-------+----+---------+
|        AAA|            2013J|     28400| 546652| -10|        4|
|        AAA|            2013J|     28400| 546652| -10|        1|
|        AAA|            2013J|     28400| 546652| -10|        1|
|        AAA|            2013J|     28400| 546614| -10|       11|
|        AAA|            2013J|     28400| 546714| -10|        1|
+-----------+-----------------+----------+-------+----+---------+
only showing top 5 rows
>>> df.count()
10655280
```

1. Big Data Environment Setup:
2. Hadoop HDFS Implementation
3. Spark data processing
4. Feature engineering

Big Data Environment Setup:

Before analyzing the dataset, the big data environment must be configured.
The project uses the Hadoop ecosystem along with Apache Spark for distributed data processing.

The setup includes installation of:

1.Hadoop

2.Apache Spark

3.Py spark

The system is configured using **WSL (Windows Subsystem for Linux)**.

After installation, Hadoop services such as **NameNode and DataNode** are started to enable distributed storage.

This environment allows large datasets to be processed efficiently.

```
arya_mulay@DARE-TO-DEVIL:~$ pyspark
Python 3.12.3 (main, Jan 22 2026, 20:57:42) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/03/08 12:53:12 WARN Utils: Your hostname, DARE-TO-DEVIL, resolves to a loopback address: 127.0.1.1; using 10.255.255.254 instead (
on interface lo)
26/03/08 12:53:12 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/03/08 12:53:15 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where a
pplicable
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 4.1.1
      /_/

Using Python version 3.12.3 (main, Jan 22 2026 20:57:42)
Spark context Web UI available at http://10.255.255.254:4040
Spark context available as 'sc' (master = local[*], app id = local-1772974397678).
SparkSession available as 'spark'.
```

Installation of Apache Spark in the WSL environment.

```
>>> full.show(50)
+----------+------------+-----------+------------+----------------+--------------------+----------+-----------+-----------+------+--------------------+--------------------+---------+---------+-----------+-----------+---------+-------------+
|id_student|total_clicks|active_days|total_events|activity_variance|            features|prediction|code_module|code_presentation|gender|              region|   highest_education|imd_band|age_band|num_of_prev_attempts|studied_credits|disability|final_result|
+----------+------------+-----------+------------+----------------+--------------------+----------+-----------+-----------+------+--------------------+--------------------+---------+---------+-----------+-----------+---------+-------------+
     25150|        2103|        122|         507|23.12561912293942[2103.0,122.0,23....|         1|        EEE|      2014B|     M|North Western Region|   HE Qualification| 90-100%|   0-35|                  0|         60|        N|         Pass|
     26247|         478|         18|         148|22.72915262575724[478.0,18.0,22.72....|         0|        FFF|      2013J|     M|   South East Region|   HE Qualification|  80-90%|  35-55|                  2|         60|        N|         Fail|
     27683|        2129|         58|         480|  71.0533257634806[2129.0,58.0,71.0....|         1|        FFF|      2014B|     M|        London Region|   HE Qualification|  30-40%|   0-35|                  0|         60|        N|         Pass|
     27780|         553|         41|         159|15.41772024652556[553.0,41.0,15.41....|         0|        EEE|      2014B|     M|               Wales|A Level or Equiva...|   10-20|   0-35|                  0|        150|        N|         Pass|
     28952|         105|         12|          45|12.69681633806958[105.0,12.0,12.60....|         0|        CCC|      2014J|     F|    Yorkshire Region|  Lower Than A Level| 60-70%|   0-35|                  0|         60|        N|    Withdrawn|
     31014|         531|         47|         176|13.22861669723932[531.0,47.0,13.22....|         0|        BBB|      2013J|     F|West Midlands Region|  Lower Than A Level|  80-90%|  35-55|                  0|        120|        N|    Withdrawn|
     31161|        2932|        185|         759|20.45192605319234[2932.0,185.0,20....|         1|        CCC|      2014J|     M|        South Region|Post Graduate Qua...| 90-100%|    55<=|                  0|        105|        N| Distinction|
     33796|         351|         45|         181| 9.609654236519363[351.0,45.0,9.609....|         0|        DDD|      2014J|     M| East Anglian Region|  Lower Than A Level|  80-90%|  35-55|                  2|         60|        N|         Pass|
     35355|        3358|        150|         839|   24.3910498126253[3358.0,150.0,24....|         1|        EEE|      2013J|     M|               Wales|  Lower Than A Level|   0-10%|  35-55|                  0|         30|        N|         Pass|
     35355|        3358|        150|         839|   24.3910498126253[3358.0,150.0,24....|         1|        CCC|      2014J|     M|               Wales|  Lower Than A Level|   0-10%|  35-55|                  0|         30|        N|    Withdrawn|
     35796|         602|         31|         194|26.23709103991063[602.0,31.0,26.23....|         0|        FFF|      2014J|     M|               Wales|A Level or Equiva...|  20-30%|   0-35|                  0|         60|        N|         Fail|
     36347|        2161|        106|         487|57.47986326568834[2161.0,106.0,57....|         1|        FFF|      2014B|     M|North Western Region|  Lower Than A Level|   10-20|   0-35|                  0|        120|        N|         Pass|
     39208|        2067|        160|         700|15.43457357196942[2067.0,160.0,15....|         1|        DDD|      2014J|     F|               Wales|A Level or Equiva...| 90-100%|  35-55|                  0|        120|        N|    Withdrawn|
     39208|        2067|        160|         700|15.43457357196942[2067.0,160.0,15....|         1|        BBB|      2014J|     F|               Wales|A Level or Equiva...| 90-100%|  35-55|                  0|        120|        N|         Pass|
     39427|          93|         14|          55|  7.5509260039073[93.0,14.0,7.5509....|         0|        DDD|      2013J|     F|        South Region|A Level or Equiva...| 90-100%|   0-35|                  0|         60|        N|    Withdrawn|
     42352|        5079|        138|        1164|47.57795887529695[5079.0,138.0,47....|         1|        FFF|      2014J|     F| East Anglian Region|A Level or Equiva...|  80-90%|   0-35|                  0|         90|        N|         Pass|
     50610|          13|          4|          11|2.872281323269014[13.0,4.0,2.87228....|         0|        DDD|      2013J|     F| East Anglian Region|A Level or Equiva...|  50-60%|   0-35|                  1|         70|        Y|         Fail|
     52130|        1894|        122|         593|15.85325994973816[1894.0,122.0,15....|         0|        AAA|      2013J|     F| East Anglian Region|A Level or Equiva...|  70-80%|   0-35|                  0|         90|        N|         Pass|
     52275|          91|          5|          27|  12.3773987574126[91.0,5.0,12.3773....|         0|        FFF|      2014J|     M|        London Region|       No Formal quals|   0-10%|   0-35|                  0|         90|        N|         Fail|
     52997|        1117|         61|         340|16.96421820932376[1117.0,61.0,16.9....|         0|        BBB|      2013J|     F|   South West Region|  Lower Than A Level| 30-40%|  35-55|                  1|         60|        N|         Pass|
     53300|        1957|        124|         669|36.5919180031926[1957.0,124.0,36....|         0|        CCC|      2014J|     M|West Midlands Region|A Level or Equiva...| 90-100%|   0-35|                  0|        270|        N| Distinction|
     55013|         604|         35|         202|20.46854529722783[604.0,35.0,20.46....|         0|        FFF|      2013J|     M|        North Region|A Level or Equiva...|   10-20|   0-35|                  0|        120|        N|         Fail|
     59272|        1425|         76|         379|26.27725823578610[1425.0,76.0,28.2....|         0|        BBB|      2014J|     F|   South East Region|  Lower Than A Level| 60-70%|   0-35|                  0|         60|        N|         Pass|
     67685|        2314|        171|         800|17.20098716377419[2314.0,171.0,17....|         1|        EEE|      2014J|     M|        South Region|A Level or Equiva...|   10-20|   0-35|                  0|        180|        N|         Pass|
     67685|        2314|        171|         800|17.20098716377419[2314.0,171.0,17....|         1|        CCC|      2014J|     M|        South Region|A Level or Equiva...|   10-20|   0-35|                  0|        180|        N|    Withdrawn|
     69439|         293|         13|         106|31.96512522686608[293.0,13.0,31.96....|         0|        FFF|      2014J|     M|East Midlands Region|  Lower Than A Level|  80-90%|   0-35|                  0|         90|        N|    Withdrawn|
     70495|        7252|        193|        1723|  40.9389255872093[7252.0,193.0,40....|         2|        FFF|      2013J|     M|            Scotland|  Lower Than A Level|  50-60%|   0-35|                  0|         60|        N| Distinction|
     70495|        7252|        193|        1723|  40.9389255872093[7252.0,193.0,40....|         2|        CCC|      2014J|     M|            Scotland|  Lower Than A Level|  50-60%|   0-35|                  0|         60|        N|         Pass|
     70569|         947|         56|         250|19.59895802955238[947.0,56.0,19.59....|         0|        BBB|      2014J|     F|               Wales|  Lower Than A Level|   0-10%|   0-35|                  3|         60|        N|         Pass|
     71433|          60|          8|          40|  4.440077219408573[60.0,8.0,4.44007....|         0|        BBB|      2013B|     F|North Western Region|A Level or Equiva...|  50-60%|   0-35|                  1|        150|        N|         Fail|
     72349|         134|         10|          54|14.90861048745545[134.0,10.0,14.90....|         0|        BBB|      2014J|     F|            Scotland|A Level or Equiva...|   0-10%|   0-35|                  0|         60|        N|    Withdrawn|
     72693|        2269|         92|         437| 53.84843287308158[2269.0,92.0,53.5....|         1|        FFF|      2013B|     F|    Yorkshire Region|  Lower Than A Level| 40-50%|   0-35|                  0|        140|        Y|         Pass|
     73584|         243|         12|          86|  27.68696775523459[243.0,12.0,27.68....|         0|        BBB|      2014J|     F|   South West Region|  Lower Than A Level| 30-40%|   0-35|                  1|         60|        N|    Withdrawn|
     77241|         224|         24|          97|   8.196853408296471[224.0,24.0,8.196....|         0|        DDD|      2013J|     M|West Midlands Region|  Lower Than A Level| 30-40%|  35-55|                  0|         60|        N|    Withdrawn|
     77704|        2273|         84|         594|32.28650028998699[2273.0,84.0,32.2....|         1|        FFF|      2013J|     M|North Western Region|A Level or Equiva...| 40-50%|   0-35|                  0|         60|        N|         Pass|
     78170|        2669|        109|         865|23.61285536520886[2669.0,109.0,23....|         1|        DDD|      2013J|     F|West Midlands Region|  Lower Than A Level| 30-40%|   0-35|                  0|         60|        N|         Fail|
     78645|         426|         40|         202|  7.677973824602559[426.0,40.0,7.677....|         0|        BBB|      2013B|     F| East Anglian Region|  Lower Than A Level| 40-50%|   0-35|                  0|         60|        N|         Fail|
     82341|         508|         29|         142|  25.71911558362653[508.0,29.0,25.71....|         0|        GGG|      2013J|     F|   South East Region|  Lower Than A Level| 90-100%|   0-35|                  0|         30|        Y| Distinction|
     82575|         658|         43|         231|22.38253521491493[658.0,43.0,22.38....|         0|        CCC|      2014B|     M|West Midlands Region|   HE Qualification|  80-90%|   0-35|                  0|         30|        N| Distinction|
     84576|        2126|        114|         702|31.34223943299282[2126.0,114.0,31....|         1|        DDD|      2014B|     M|               Wales|   HE Qualification| 30-40%|  35-55|                  0|        150|        N| Distinction|
     84604|        7698|        156|        1645|49.31473440431016[7698.0,156.0,49....|         2|        FFF|      2014J|     F|        North Region|A Level or Equiva...| 60-70%|  35-55|                  0|         90|        N| Distinction|
     84812|         474|         48|         181|10.43440582018948[474.0,48.0,10.43....|         0|        BBB|      2013J|     F| East Anglian Region|A Level or Equiva...|  80-90%|   0-35|                  0|        120|        N|         Pass|
     85241|         204|         16|          93|  11.34019408186787[204.0,16.0,11.34....|         0|        FFF|      2014J|     M|   South West Region|  Lower Than A Level| 50-60%|   0-35|                  0|        120|        Y|    Withdrawn|
     90243|        1084|         57|         321|  17.78891238191153[1084.0,57.0,17.7....|         0|        BBB|      2013J|     F|               Wales|   HE Qualification| 70-80%|  35-55|                  0|         60|        N|         Pass|
     92341|         957|         54|         270|20.64106245132405[957.0,54.0,20.64....|         0|        BBB|      2014B|     F|North Western Region|  Lower Than A Level| 40-50%|   0-35|                  2|         60|        N|         Fail|
     92947|        1350|        129|         494|15.10796898144108[1350.0,129.0,15....|         0|        BBB|      2013B|     F|               Wales|   HE Qualification|  10-20|  35-55|                  0|        120|        N| Distinction|
     94961|        1240|         93|         417|16.45767518405221[1240.0,93.0,16.4....|         0|        AAA|      2014J|     M|        South Region|  Lower Than A Level| 70-80%|  35-55|                  1|         60|        N|         Pass|
     94961|        1240|         93|         417|16.45767518405221[1240.0,93.0,16.4....|         0|        AAA|      2013J|     M|        South Region|  Lower Than A Level| 70-80%|  35-55|                  0|         60|        N|    Withdrawn|
     95755|         381|         43|         127|11.57908891066793[381.0,43.0,11.57....|         0|        AAA|      2014J|     M|   South East Region|  Lower Than A Level| 90-100%|   0-35|                  0|         60|        N|    Withdrawn|
     96873|         402|         22|         127|30.36787434457328[402.0,22.0,30.36....|         0|        EEE|      2013J|     M|North Western Region|A Level or Equiva...| 70-80%|   0-35|                  0|         30|        N|         Fail|
+----------+------------+-----------+------------+----------------+--------------------+----------+-----------+-----------+------+--------------------+--------------------+---------+---------+-----------+-----------+---------+-------------+
```

Configuration of the big data processing environment.

Hadoop HDFS Implementation:

The Hadoop Distributed File System (HDFS) is used to store the OULAD dataset.

HDFS stores data across multiple nodes, which enables large datasets to be processed efficiently. The dataset is uploaded into HDFS using the following command:

hdfs dfs -put /local_path/*.csv /oulad

This command uploads all CSV dataset files into the **/oulad directory in HDFS**. Once uploaded, the dataset can be accessed by Spark for analysis.

```
arya_mulay@DARE-TO-DEVIL:~$ ls ~/hadoopdata/hdfs/namenode
current
arya_mulay@DARE-TO-DEVIL:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [DARE-TO-DEVIL]
jpsarya_mulay@DARE-TO-DEVIL:~$ jps
2753 DataNode
2994 SecondaryNameNode
3141 Jps
2613 NameNode
arya_mulay@DARE-TO-DEVIL:~$ hdfs dfs -mkdir /oulad
arya_mulay@DARE-TO-DEVIL:~$ ls ~/oulad
assessments.csv        studentInfo.csv          vle.csv
courses.csv            studentRegistration.csv
studentAssessment.csv  studentVle.csv
arya_mulay@DARE-TO-DEVIL:~$ hdfs dfs -put ~/oulad/studentVle.csv /oulad/
arya_mulay@DARE-TO-DEVIL:~$ hdfs dfs -ls /oulad
Found 1 items
-rw-r--r--   1 arya_mulay supergroup  453836331 2026-04-04 03:04 /oulad/studentVle.csv
```

Uploading dataset files into Hadoop HDFS and verifying

Spark Data Processing:

Apache Spark is used for processing the dataset after it is stored in HDFS.

Spark performs distributed computations and processes large datasets efficiently.

Spark reads the dataset from HDFS and performs the following operations:

1. Data cleaning.
2. Aggregation of student clicks
3. Preparation of features for machine learning

Spark is faster than traditional processing systems because it uses **in-memory computation**.

Cluster	Interpretation
0	Active learners
1	Moderate learners
2	Learning collapse risk

Low clicks

Few active days

Unstable activity

Processing the dataset using Apache Spark.

Feature Engineering:

Feature engineering is the process of converting raw student activity data into meaningful behavioral features.

In this project, features are generated from click

stream activity logs

Examples of features include:

1. Total clicks per student

2. Average clicks per day

3. Variation in engagement levels

4. Activity frequency

These features help understand how actively students participate in online

learning. Students whose engagement decreases over time may be

experiencing learning collapse.

```
daily = df.groupBy("aggregating every student",
"date").agg( sum("sum_click").alias("daily_clicks")

)

from pyspark.sql.functions import stddev instability
= daily.groupBy("id_student").agg(

stddev("daily_clicks").alias("activity_variance")

)

dataset = features.join(instability, "id_student")
dataset.show()
```

id_student	total_clicks	active_days	total_events	activity_variance
25150	2103	122	507	23.125619322933428
26247	478	18	148	22.729152625757244
27683	2129	50	400	71.05332576340685
27780	553	41	159	15.417720245255962
28952	105	12	45	12.606816338069583
31014	531	47	176	13.228616697239328
31161	2932	185	759	20.451926953192142
33796	351	45	181	9.609654236519363
35355	3358	150	839	24.3910498126253
35796	602	31	194	26.237091039910638
36347	2161	106	487	57.479863265688344
39208	2067	160	700	15.434573571069421
39427	93	14	55	7.550926003907303
42552	5079	138	1164	47.57795887529695
50610	13	4	11	2.8722813232690143
52130	1894	122	593	15.853259494738166
52275	91	5	27	12.37739875741264
52797	1117	61	340	16.96421820932376
53300	1957	124	669	36.59191800319264
55013	604	35	202	20.468545247227883

only showing top 20 rows

Output of behavioral feature extraction.

What this step actually did (important for review explanation)

You created **behavior features for each student**.

Feature	Meaning
total_clicks	total interactions with learning materials
active_days	number of days the student studied
total_events	number of learning resources accessed

These represent **student engagement levels**.

Feature extraction process for behavioral indicators.

```
>>> features.show()
+----------+------------+-----------+------------+
|id_student|total_clicks|active_days|total_events|
+----------+------------+-----------+------------+
|    180753|        2012|        157|         858|
|    486656|         757|         54|         299|
|    324084|        1961|        110|         494|
|   2057803|        3418|        147|         862|
|    537558|         658|         76|         297|
|    502604|         328|         44|         150|
|    578600|        1344|         90|         404|
|    555673|        3550|         92|         692|
|    483876|         417|         40|         213|
|     84812|         474|         48|         181|
|    597532|         401|         36|         174|
|    633208|         473|         65|         228|
|    653958|         991|         57|         299|
|    635591|         688|         57|         302|
|   1840034|         376|         50|         199|
|    605068|        1854|        106|         474|
|    608407|        1453|        104|         534|
|    689665|        2152|         63|         416|
|    607909|         253|         25|         116|
|    535379|         377|         29|         138|
+----------+------------+-----------+------------+
only showing top 20 rows
>>>
```

Final behavioral feature dataset.

Machine Learning Model:

The project uses KMeans clustering to analyze student engagement patterns.

KMeans is an unsupervised machine learning algorithm that groups similar data points into clusters.

Students are grouped based on their engagement features.

Example clusters:

Cluster 1(0)→ Highly active students

Cluster 2(1)→ Moderately active students

Cluster 3(2)→ Low engagement students

```
from pyspark.ml.feature import VectorAssembler assembler = VectorAssembler(
inputCols=["total_clicks", "active_days", "activity_variance"], outputCol="features"
)
data = assembler.transform(dataset)
from pyspark.ml.clustering import KMeans kmeans = KMeans(k=3)
model = kmeans.fit(data) predictions = model.transform(data) predictions.show()
```

Implementation of the KMeans clustering algorithm.

```
>>> predictions.show()
+----------+------------+-----------+------------+------------------+--------------------+----------+
|id_student|total_clicks|active_days|total_events| activity_variance|            features|prediction|
+----------+------------+-----------+------------+------------------+--------------------+----------+
|     25150|        2103|        122|         507|23.125619322933428|[2103.0,122.0,23....|         1|
|     26247|         478|         18|         148|22.729152625757244|[478.0,18.0,22.72...|         0|
|     27683|        2129|         50|         400| 71.05332576340685|[2129.0,50.0,71.0...|         1|
|     27780|         553|         41|         159|15.417720245255962|[553.0,41.0,15.41...|         0|
|     28952|         105|         12|          45|12.606816338069583|[105.0,12.0,12.60...|         0|
|     31014|         531|         47|         176|13.228616697239328|[531.0,47.0,13.22...|         0|
|     31161|        2932|        185|         759|20.451926953192142|[2932.0,185.0,20....|         1|
|     33796|         351|         45|         181| 9.609654236519363|[351.0,45.0,9.609...|         0|
|     35355|        3358|        150|         839|  24.3910498126253|[3358.0,150.0,24....|         1|
|     35796|         602|         31|         194|26.237091039910638|[602.0,31.0,26.23...|         0|
|     36347|        2161|        106|         487|57.479863265688344|[2161.0,106.0,57....|         1|
|     39208|        2067|        160|         700|15.434573571069421|[2067.0,160.0,15....|         1|
|     39427|          93|         14|          55| 7.550926003907303|[93.0,14.0,7.5509...|         0|
|     42552|        5079|        138|        1164| 47.57795887529695|[5079.0,138.0,47....|         1|
|     50610|          13|          4|          11|2.8722813232690143|[13.0,4.0,2.87228...|         0|
|     52130|        1894|        122|         593|15.853259494738166|[1894.0,122.0,15....|         0|
|     52275|          91|          5|          27| 12.37739875741264|[91.0,5.0,12.3773...|         0|
|     52797|        1117|         61|         340| 16.96421820932376|[1117.0,61.0,16.9...|         0|
|     53300|        1957|        124|         669| 36.59191800319264|[1957.0,124.0,36....|         0|
|     55013|         604|         35|         202|20.468545247227883|[604.0,35.0,20.46...|         0|
+----------+------------+-----------+------------+------------------+--------------------+----------+
only showing top 20 rows
```

Clustering output showing student engagement groups.

Visuaization and Analysis:

Visualization plays an important role in understanding the results produced by the machine learning model. After applying clustering on the student engagement data, it becomes easier to interpret the patterns using graphical representations.

In this project, visualization is performed using Python libraries such as Matplotlib and Seaborn. These libraries help convert numerical data into graphs and charts that are easy to understand.

The visualizations help in analyzing several aspects of student learning behavior.

First, the graphs show how student engagement is distributed across the dataset. Some students interact with the learning platform very frequently, while others show very limited activity. Visualizing this distribution helps in understanding the overall participation level of students in the course.

Second, visualization helps in understanding the clusters generated by the KMeans algorithm. After clustering is performed, students are grouped into different categories based on their engagement patterns. These clusters can be visualized using charts that show how many students belong to each cluster.

For example:

1. Cluster 1 may represent highly active students

2. Cluster 2 may represent moderately active students

3. Cluster 3 may represent low engagement students

These clusters help identify groups of students who may require additional support.

Third, visualization helps identify learning behavior trends over time. By analyzing activity patterns across days or weeks, it becomes possible to observe whether student engagement is increasing, decreasing, or remaining stable.

If a student's activity gradually decreases over time, it may indicate learning instability or possible learning collapse.

Visualization therefore helps in making the analysis more understandable and supports better interpretation of the machine learning results.

Overall, visualization provides a clear view of the learning behavior of students and helps educators easily identify patterns in student engagement.

```
>>> predictions.show()
+----------+------------+-----------+------------+------------------+--------------------+----------+
|id_student|total_clicks|active_days|total_events| activity_variance|            features|prediction|
+----------+------------+-----------+------------+------------------+--------------------+----------+
|     25150|        2103|        122|         507|23.125619322933428|[2103.0,122.0,23....|         1|
|     26247|         478|         18|         148|22.729152625757244|[478.0,18.0,22.72...|         0|
|     27683|        2129|         50|         400| 71.05332576340685|[2129.0,50.0,71.0...|         1|
|     27780|         553|         41|         159|15.417720245255962|[553.0,41.0,15.41...|         0|
|     28952|         105|         12|          45|12.606816338069583|[105.0,12.0,12.60...|         0|
|     31014|         531|         47|         176|13.228616697239328|[531.0,47.0,13.22...|         0|
|     31161|        2932|        185|         759|20.451926953192142|[2932.0,185.0,20....|         1|
|     33796|         351|         45|         181| 9.609654236519363|[351.0,45.0,9.609...|         0|
|     35355|        3358|        150|         839|  24.3910498126253|[3358.0,150.0,24....|         1|
|     35796|         602|         31|         194|26.237091039910638|[602.0,31.0,26.23...|         0|
|     36347|        2161|        106|         487|57.479863265688344|[2161.0,106.0,57....|         1|
|     39208|        2067|        160|         700|15.434573571069421|[2067.0,160.0,15....|         1|
|     39427|          93|         14|          55| 7.550926003907303|[93.0,14.0,7.5509...|         0|
|     42552|        5079|        138|        1164| 47.57795887529695|[5079.0,138.0,47....|         1|
|     50610|          13|          4|          11|2.8722813232690143|[13.0,4.0,2.87228...|         0|
|     52130|        1894|        122|         593|15.853259494738166|[1894.0,122.0,15....|         0|
|     52275|          91|          5|          27| 12.37739875741264|[91.0,5.0,12.3773...|         0|
|     52797|        1117|         61|         340| 16.96421820932376|[1117.0,61.0,16.9...|         0|
|     53300|        1957|        124|         669| 36.59191800319264|[1957.0,124.0,36....|         0|
|     55013|         604|         35|         202|20.468545247227883|[604.0,35.0,20.46...|         0|
+----------+------------+-----------+------------+------------------+--------------------+----------+
only showing top 20 rows
```

Visualization of student clusters based on engagement behavior.

```
ers > venka > AppData > Local > Temp > 9245bbad-0305-4c38-a166-a55c1ef4243e_test.zip.43e > test > clickstream_data_rows.csv
id_student,id_site,date,sum_click
23f18d22-3bd2-4885-810e-1adc7140af16,assessments,1,2
23f18d22-3bd2-4885-810e-1adc7140af16,assessments_910644c9-3cd6-4477-a19a-abfff2640f74,1,34
23f18d22-3bd2-4885-810e-1adc7140af16,course,1,6
23f18d22-3bd2-4885-810e-1adc7140af16,dashboard,1,9
23f18d22-3bd2-4885-810e-1adc7140af16,exam,1,2
23f18d22-3bd2-4885-810e-1adc7140af16,home,1,2
23f18d22-3bd2-4885-810e-1adc7140af16,login,1,17
2f4327d3-cf9d-408c-9b65-a6c52e1c8f5c,assessments_0039135c-8033-4312-8904-2b415522edc6,1,13
2f4327d3-cf9d-408c-9b65-a6c52e1c8f5c,assessments_910644c9-3cd6-4477-a19a-abfff2640f74,1,10
2f4327d3-cf9d-408c-9b65-a6c52e1c8f5c,course,1,11
2f4327d3-cf9d-408c-9b65-a6c52e1c8f5c,dashboard,1,1
2f4327d3-cf9d-408c-9b65-a6c52e1c8f5c,login,1,1
62d4fd0e-ab93-4b20-90bd-cf0e8c75057b,dashboard,1,5
62d4fd0e-ab93-4b20-90bd-cf0e8c75057b,home,1,2
62d4fd0e-ab93-4b20-90bd-cf0e8c75057b,login,1,4
697394ad-3f76-4907-99ef-fa125437031d,assessments,1,11
697394ad-3f76-4907-99ef-fa125437031d,assessments_910644c9-3cd6-4477-a19a-abfff2640f74,1,10
697394ad-3f76-4907-99ef-fa125437031d,course,1,6
697394ad-3f76-4907-99ef-fa125437031d,dashboard,1,12
697394ad-3f76-4907-99ef-fa125437031d,exam,1,13
697394ad-3f76-4907-99ef-fa125437031d,login,1,1
6fe3dc55-b1d5-4c89-b9bb-d8e4a15a3453,assessments_910644c9-3cd6-4477-a19a-abfff2640f74,1,12
6fe3dc55-b1d5-4c89-b9bb-d8e4a15a3453,course,1,8
6fe3dc55-b1d5-4c89-b9bb-d8e4a15a3453,dashboard,1,1
6fe3dc55-b1d5-4c89-b9bb-d8e4a15a3453,login,1,1
6fe3dc55-b1d5-4c89-b9bb-d8e4a15a3453,week,1,9
75ae17af-97e6-48ec-b20d-0d8f44aff880,assessments,1,4
75ae17af-97e6-48ec-b20d-0d8f44aff880,assessments_910644c9-3cd6-4477-a19a-abfff2640f74,1,9
75ae17af-97e6-48ec-b20d-0d8f44aff880,course,1,6
75ae17af-97e6-48ec-b20d-0d8f44aff880,dashboard,1,7
75ae17af-97e6-48ec-b20d-0d8f44aff880,exam,1,7
75ae17af-97e6-48ec-b20d-0d8f44aff880,home,1,4
75ae17af-97e6-48ec-b20d-0d8f44aff880,login,1,4
75ae17af-97e6-48ec-b20d-0d8f44aff880,week,1,2
78b0353a-c989-4477-a78b-37a4ee035dd4,assessments,1,3
78b0353a-c989-4477-a78b-37a4ee035dd4,course,1,4
```

Click_Stream_Data_row

6. Results and Discussion:

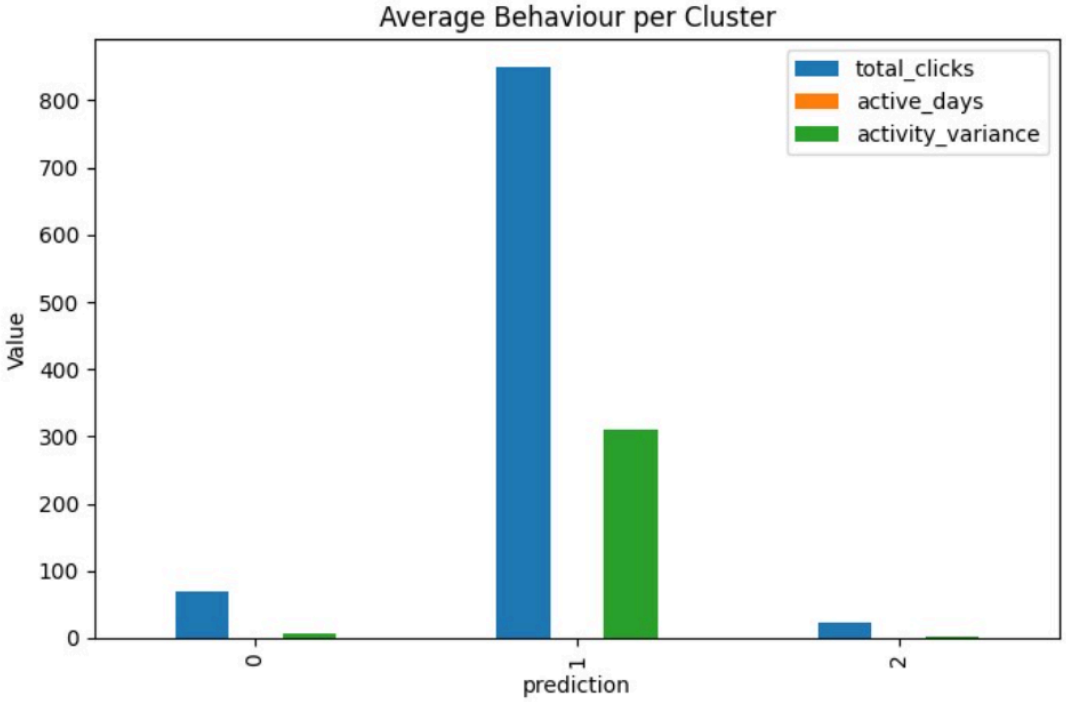
The results of this project show that big data technologies can be effectively used to analyze large volumes of student activity data and understand learning behavior. In this system, Hadoop HDFS is used to store the dataset in a distributed manner, while tools like Apache Spark are used to process and analyze the data efficiently. After processing, important behavioral features such as total clicks, active days, and engagement variation are generated, which help in understanding how students interact with the learning platform over time.

Using these features, the K-Means clustering algorithm groups students based on their engagement levels. The results clearly show different types of learners: some students are highly active and regularly interact with learning materials, while others show moderate participation. At the same time, a group of students shows very low or inconsistent activity. These students are identified as being at risk of learning collapse because their learning behavior is unstable or declining, even if they are still enrolled in the course.

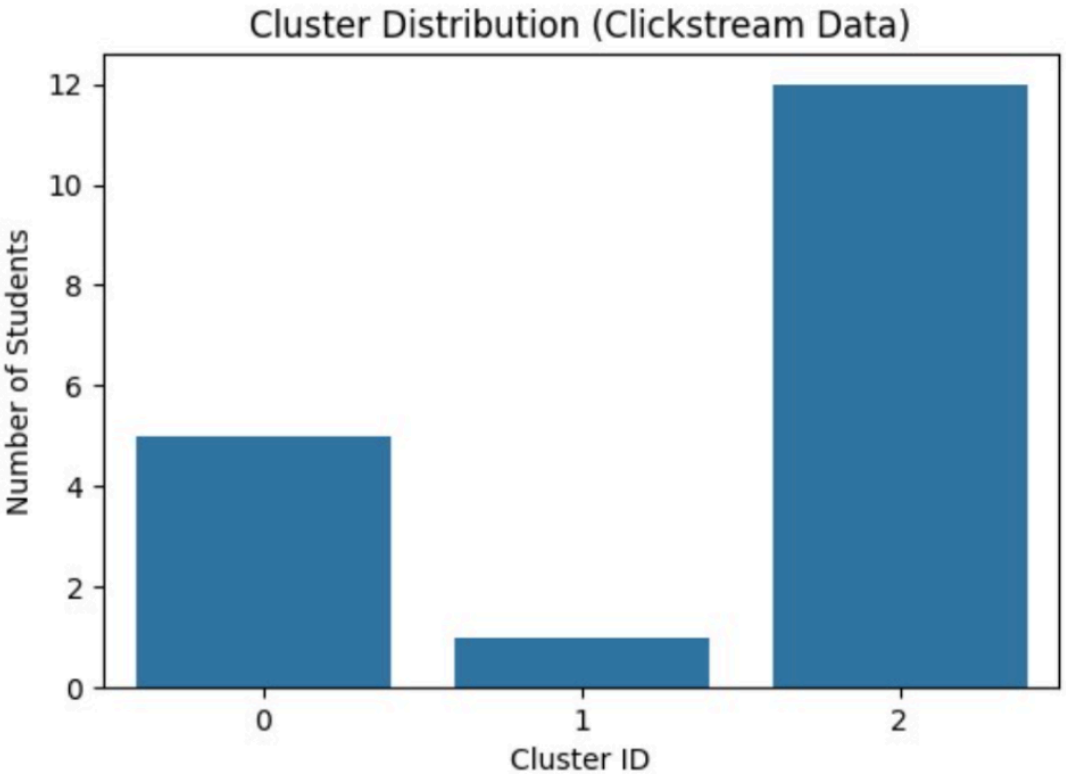
The analysis highlights that student engagement is not the same for everyone and that behavioral patterns can reveal hidden learning problems before academic failure occurs. By identifying these patterns early, educators can provide timely support such as additional guidance, personalized learning strategies, or intervention programs. Overall, the results demonstrate that big data analytics combined with machine learning can play an important role in detecting early learning issues and improving student success.

```
sers > venka > AppData > Local > Temp > 8b33ac17-fc2e-456f-b9eb-2fb45a89a190_test.zip.190 > test > final_clickstream_output.csv
id_student,total_clicks,active_days,total_events,activity_variance,prediction
62d4fd0e-ab93-4b20-90bd-cf0e8c75057b,11,1,3,1.5275252316519468,2
7b43790a-6efa-4a0f-9be7-8e8fd1dd1108,25,1,5,2.23606797749979,2
d74a85bf-a23e-4c51-beaf-68d78ac2c74d,4,1,3,0.5773502691896258,2
75ae17af-97e6-48ec-b20d-0d8f44aff880,43,1,8,2.263846284534354,2
df14f4a7-f8a1-4a9c-8247-67dddcf97688,25,1,5,2.7386127875258306,2
8229c005-8752-4e52-a513-86eb4dc7587b,849,1,7,310.3201112215762,1
7d5aaad6-7b09-43ca-b66f-8997ec76139f,77,1,8,6.8855438627398415,0
a5ca9730-27c1-47b7-a1f5-c90edd0f0498,36,1,6,3.7416573867739413,2
df2a586b-ed41-419c-b6fc-ef3550fff1d2,11,1,2,2.1213203435596424,2
8fc1b351-9614-4a41-9ba4-92f5f7920d3a,99,1,11,4.753945729601885,0
2f4327d3-cf9d-408c-9b65-a6c52e1c8f5c,36,1,5,5.761944116355173,2
6fe3dc55-b1d5-4c89-b9bb-d8e4a15a3453,31,1,5,4.969909455915671,2
a134d98d-d1fd-4d85-9767-af693e9a513d,52,1,7,4.07664661444276,0
23f18d22-3bd2-4885-810e-1adc7140af16,72,1,7,11.785786435565592,0
697394ad-3f76-4907-99ef-fa125437031d,53,1,6,4.535048695071164,0
d1848ddf-2d89-4439-91c7-7af8f3f624ac,14,1,4,2.380476142847616,2
78b0353a-c989-4477-a78b-37a4ee035dd4,11,1,4,1.2583057392117918,2
db562bf4-3643-48a8-bc94-09b82accf0d9,29,1,4,3.5000000000000004,2
```

Final click stream output



Final Data Visualization Graph



Final cluster Distribution Graph

7. Conclusion:

This project demonstrates how big data technologies and machine learning techniques can be used to analyze student activity patterns and detect learning collapse at an early stage. By using the Open University Learning Analytics Dataset, the system studies how students interact with learning materials and how their engagement changes over time. Technologies such as Hadoop HDFS and Apache Spark are used to store and process large-scale educational data efficiently.

Through feature engineering, meaningful behavioral indicators like total clicks, active days, and engagement variation are extracted from raw data. These features help in understanding student learning behavior more clearly. The application of the K-Means clustering algorithm groups students based on their engagement levels, making it easier to identify students who are highly active, moderately active, or at risk of learning collapse. The results show that students with low or unstable engagement patterns can be detected before actual academic failure occurs.

Overall, this project highlights the importance of analyzing learning behavior instead of only focusing on final outcomes. It proves that early detection of learning problems using data-driven approaches can help educators provide timely support and improve student performance.

Future Work:

In the future, this work can be improved by using real-time data processing instead of only analyzing historical data. This will help in detecting learning collapse as it happens and allow faster intervention. More advanced machine learning models such as deep learning or time-series analysis can be used to improve accuracy and capture complex learning patterns over time.

The system can also be enhanced by integrating additional data sources such as student feedback, attendance, and assessment performance for a more complete analysis. Another improvement could be the development of a real-time dashboard that provides alerts to teachers when a student shows signs of learning collapse.

Furthermore, personalized recommendation systems can be added to suggest learning resources or strategies based on each student's behavior. Expanding the system to work across different educational platforms and larger datasets will also make it more scalable and practical for real-world use.

Important Links:

Complete Project Git-Hub repository: {Click Here}

Vercel Deployment for the web simulation data: {Click Here}

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